Anexo I

Regras das derivadas, considere *c* constante e *u*, *v* são variáveis de *x*.

I. Constante: $\frac{\partial c}{\partial x} = 0$
II. Produto por uma constante: $\frac{\partial c.u}{\partial x} = c.\frac{\partial u}{\partial x}$
III. Derivada do produto: $\frac{\partial (u.v)}{\partial x} = u.\frac{\partial v}{\partial x} + v.\frac{\partial u}{\partial x}$
IV. Derivada da subtração: $\frac{\partial (u-v)}{\partial x} = \frac{\partial u}{\partial x} - \frac{\partial v}{\partial x}$
V. Derivada da soma: $\frac{\partial (u+v)}{\partial x} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial x}$

Resolvendo:

$$\frac{\partial E_{TOTAL}}{\partial g(o_1)} = \frac{1}{2}.(d_1 - g(o_1))^2 + \frac{1}{2}.(d_2 - g(o_2))^2$$

Aplique a regra V na equação.
Então temos:

$$\frac{\partial E_{TOTAL}}{\partial g(o_1)} = \frac{\partial \frac{1}{2}.(d_1 - g(o_1))^2}{\partial g(o_1)} + \frac{\partial \frac{1}{2}.(d_2 - g(o_2))^2}{\partial g(o_1)}$$

Aplique a regra I, dessa forma, esta parte $\frac{\partial \frac{1}{2}.(d_2 - g(o_2))^2}{\partial g(o_1)} = 0$

$$\frac{\partial E_{TOTAL}}{\partial g(o_1)} = \frac{\partial \frac{1}{2}.(d_1 - g(o_1))^2}{\partial g(o_1)} + 0$$

Aplico a regra II.

$$\frac{\partial E_{TOTAL}}{\partial g(o_1)} = \frac{\partial \frac{1}{2}.(d_1 - g(o_1))^2}{\partial g(o_1)} = \frac{\frac{1}{2}\partial (d_1 - g(o_1))^2}{\partial g(o_1)} = \frac{\frac{1}{2}\partial ((d_1 - g(o_1))(d_1 - g(o_1)))}{\partial g(o_1)}$$

Aplico a regra III

$$\frac{\partial E_{TOTAL}}{\partial g(o_1)} = \frac{\frac{1}{2}\partial ((d_1 - g(o_1))(d_1 - g(o_1)))}{\partial g(o_1)}$$

$$= \frac{\frac{1}{2}\{((d_1 - g(o_1))(\partial (d_1 - g(o_1))) + (\partial (d_1 - g(o_1)))(d_1 - g(o_1)))\}}{\partial g(o_1)}$$

Essa parcela tem este valor: $\frac{(\partial (d_1 - g(o_1)))}{\partial g(o_1)} = -g_{o_1}^{1-1} = -1g(o_1)^0 = -1$pois aplicamos a regra V e I.

Rescrevendo

$$\frac{\partial E_{TOTAL}}{\partial g(o_1)} = \frac{1}{2}\{(d_1 - g(o_1))(-1) + (-1)(d_1 - g(o_1))\} = \frac{1}{2}\{-(d_1 - g(o_1)) + (-(d_1 - g(o_1)))\}$$

$$\frac{\partial E_{TOTAL}}{\partial g(o_1)} = \frac{1}{2}\{-(d_1 - g(o_1)) - (d_1 - g(o_1))\} = \frac{1}{2}\{-2(d_1 - g(o_1)) = \frac{-2}{2}(d_1 - g(o_1))$$

$$= -(d_1 - g(o_1))$$

Anexo II

Regras das derivadas, considere *c* constante e *u*, *v* são variáveis de *x*.

I. Constante: $\frac{\partial c}{\partial x} = 0$

II. Produto por uma constante: $\frac{\partial c.u}{\partial x} = c.\frac{\partial u}{\partial x}$
III. Derivada do produto: $\frac{\partial (u.v)}{\partial x} = u.\frac{\partial v}{\partial x} + v.\frac{\partial u}{\partial x}$
IV. Derivada da subtração: $\frac{\partial (u-v)}{\partial x} = \frac{\partial u}{\partial x} - \frac{\partial v}{\partial x}$
V. Derivada da soma: $\frac{\partial (u+v)}{\partial x} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial x}$
VI. Derivada da divisão: $\frac{\partial (\frac{u}{v})}{\partial x} = \frac{v\frac{\partial u}{\partial x} - u\frac{\partial v}{\partial x}}{v^2}$

Quero demostrar que:
$$\frac{\partial g(o_1)}{\partial NET_{o_1}} = g(o_1).(1 - g(o_1))$$

Resolvendo:
$$\frac{\partial g(o_1)}{\partial NET_{o_1}} = \frac{\partial \frac{1}{1 + e^{-NET_{o_1}}}}{\partial NET_{o_1}}$$
Aplico a regra VI
$$\frac{\partial g(o_1)}{\partial NET_{o_1}} = \frac{\partial \frac{1}{1 + e^{-NET_{o_1}}}}{\partial NET_{o_1}} = \frac{(1 + e^{-NET_{o_1}}).\frac{\partial 1}{\partial NET_{o_1}} - (1).\frac{\partial [1 + e^{-NET_{o_1}}]}{\partial NET_{o_1}}}{(1 + e^{-NET_{o_1}})^2}$$
$$= \frac{(1 + e^{-NET_{o_1}}).0 - (1).[\frac{\partial 1}{\partial NET_{o_1}} + \frac{\partial e^{-NET_{o_1}}}{\partial NET_{o_1}}]}{(1 + e^{-NET_{o_1}})^2}$$

$$\frac{\partial g(o_1)}{\partial NET_{o_1}} = \frac{0 - (1).[0 + \frac{\partial e^{-NET_{o_1}}]}{\partial NET_{o_1}}}{(1 + e^{-NET_{o_1}})^2} = \frac{-[-e^{-NET_{o_1}}.(1)]}{(1 + e^{-NET_{o_1}})^2} = \frac{e^{-NET_{o_1}}}{(1 + e^{-NET_{o_1}})^2}$$
$$= \frac{e^{-NET_{o_1}}}{(1 + e^{-NET_{o_1}}).(1 + e^{-NET_{o_1}})}$$

$$\frac{\partial g(o_1)}{\partial NET_{o_1}} = \frac{e^{-NET_{o_1}}}{(1 + e^{-NET_{o_1}})}.\frac{1}{(1 + e^{-NET_{o_1}})} = \frac{1 - 1 + e^{-NET_{o_1}}}{(1 + e^{-NET_{o_1}})}.\frac{1}{(1 + e^{-NET_{o_1}})}$$
$$= \frac{1 + e^{-NET_{o_1}} - 1}{(1 + e^{-NET_{o_1}})}.\frac{1}{(1 + e^{-NET_{o_1}})}$$

$\frac{\partial g(o_1)}{\partial NET_{o_1}} = [\frac{1+e^{-NET_{o_1}}}{(1+e^{-NET_{o_1}})} - \frac{1}{(1+e^{-NET_{o_1}})}].\frac{1}{(1+e^{-NET_{o_1}})} = [1 - \frac{1}{(1+e^{-NET_{o_1}})}].\frac{1}{(1+e^{-NET_{o_1}})}.$
Devemos lembrar que
$g(o_1) = \frac{1}{1+e^{-NET_{o_1}}}$ então temos:

$$\frac{\partial g(o_1)}{\partial NET_{o_1}} = [1 - \frac{1}{(1 + e^{-NET_{o_1}})}].\frac{1}{(1 + e^{-NET_{o_1}})} = (1 - g(o_1)).g(o_1) = g(o_1).(1 - g(o_1))$$

Anexo III

Regras das derivadas, considere *c* constante e *u*, *v* são variáveis de *x*.

I. Constante: $\frac{\partial c}{\partial x} = 0$
II. Produto por uma constante: $\frac{\partial c.u}{\partial x} = c.\frac{\partial u}{\partial x}$
III. Derivada do produto: $\frac{\partial (u.v)}{\partial x} = u.\frac{\partial v}{\partial x} + v.\frac{\partial u}{\partial x}$
IV. Derivada da subtração: $\frac{\partial (u-v)}{\partial x} = \frac{\partial u}{\partial x} - \frac{\partial v}{\partial x}$
V. Derivada da soma: $\frac{\partial (u+v)}{\partial x} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial x}$
VI. Derivada da divisão: $\frac{\partial (\frac{u}{v})}{\partial x} = \frac{v\frac{\partial u}{\partial x} - u\frac{\partial v}{\partial x}}{v^2}$

Quero demostrar que:
$$\frac{\partial NET_{o_1}}{\partial w_5} = \frac{\partial [g(h_1).w_5 + g(h_2).w_6 + b_2.1]}{\partial w_5} = g(h_1)$$

Temos que
$$\frac{\partial NET_{o_1}}{\partial w_5} = \frac{\partial [g(h_1).w_5 + g(h_2).w_6 + b_2.1]}{\partial w_5} = \frac{\partial (g(h_1).w_5)}{\partial w_5} + \frac{\partial (g(h_2).w_6)}{\partial w_5} + \frac{\partial (b_2.1)}{\partial w_5}$$
Após aplicada a regra V, utilizamos a regra III.
$$\frac{\partial NET_{o_1}}{\partial w_5} = [g(h_1).(1) + 0.w_5] + [g(h_2).0 + 0.w_6] + [b_2.0 + 0.b_2] = g(h_1)$$

Anexo IV

Regras das derivadas, considere *c* constante e *u*, *v* são variáveis de *x*.

I. Constante: $\frac{\partial c}{\partial x} = 0$
II. Produto por uma constante: $\frac{\partial c.u}{\partial x} = c.\frac{\partial u}{\partial x}$
III. Derivada do produto: $\frac{\partial (u.v)}{\partial x} = u.\frac{\partial v}{\partial x} + v.\frac{\partial u}{\partial x}$
IV. Derivada da subtração: $\frac{\partial (u-v)}{\partial x} = \frac{\partial u}{\partial x} - \frac{\partial v}{\partial x}$

V. Derivada da soma: $\frac{\partial(u+v)}{\partial x} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial x}$

VI. Derivada da divisão: $\frac{\partial(\frac{u}{v})}{\partial x} = \frac{v\frac{\partial u}{\partial x} - u\frac{\partial v}{\partial x}}{v^2}$

Queremos mostrar que $\frac{\partial E_{o_1}}{\partial g(o_1)} = \frac{\partial[\frac{1}{2}.(d_1 - g(o_1))^2]}{\partial g(o_1)} = (g(o_1) - d_1).$

Aplique a regra II

$$\frac{\partial E_{o_1}}{\partial g(o_1)} = \frac{\partial[\frac{1}{2}.(d_1 - g(o_1))^2]}{\partial g(o_1)} = \frac{\frac{1}{2}\partial[(d_1 - g(o_1))^2]}{\partial g(o_1)} = \frac{\frac{1}{2}\partial[(d_1 - g(o_1)).(d_1 - g(o_1))]}{\partial g(o_1)}$$

Aplique a regra III

$$\frac{\partial E_{o_1}}{\partial g(o_1)} = \frac{\frac{1}{2}[(d_1 - g(o_1)).\partial(d_1 - g(o_1)) + (d_1 - g(o_1)).\partial(d_1 - g(o_1))]}{\partial g(o_1)}$$

Aplique a regra IV

$$\frac{\partial E_{o_1}}{\partial g(o_1)} = \frac{\frac{1}{2}[(d_1 - g(o_1)).(\partial d_1 - \partial g(o_1)) + (d_1 - g(o_1)).(\partial d_1 - \partial g(o_1))]}{\partial g(o_1)}$$

$$\frac{\partial E_{o_1}}{\partial g(o_1)} = \frac{1}{2}[(d_1 - g(o_1)).(0 - 1) + (d_1 - g(o_1)).(0 - 1)]$$

$$= \frac{1}{2}[-(d_1 - g(o_1)) - (d_1 - g(o_1))]$$

$$\frac{\partial E_{o_1}}{\partial g(o_1)} = \frac{1}{2}[-2(d_1 - g(o_1))] = \frac{1}{2}[2(-d_1 + g(o_1))] = (-d_1 + g(o_1)) = g(o_1) - d_1$$